

ARPAN RAJ

Cloud & DevOps | AWS Infrastructure | Software Engineer

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- **Live AWS Self-Healing Infrastructure:** Intentionally terminate a production ECS task and observe automatic service recovery.

Highlights

- **2 years of full-time production engineering experience** with a New York-based company
- Designed and operated **production AWS infrastructure** for a live e-commerce platform
- Hands-on experience with **ECS Fargate, ECR, ALB, CloudFront, Route 53, S3, Lambda and CloudWatch**
- Built **automated CI/CD pipelines** using Docker, Jenkins and GitHub Actions
- Built a **live AWS self-healing infrastructure platform** demonstrating automated ECS task recovery
- End-to-end ownership across **software development, cloud infrastructure, deployment and production operations**

PROFESSIONAL SUMMARY

Cloud & Software Engineer with ~2 years of full-time production engineering experience, specializing in AWS infrastructure, cloud-native applications, CI/CD and full-stack development. Worked as the lead engineer responsible for designing, building, deploying and operating the production technology stack for a New York-based e-commerce company. Hands-on experience across **AWS, Docker, CI/CD, Linux, React, Node.js and MongoDB**, with end-to-end ownership from application development through cloud deployment and production operations.

TECHNICAL SKILLS

Cloud & Infrastructure:

AWS (ECS Fargate, ECR, EC2, S3, CloudFront, Lambda, IAM, Route 53, ALB, CloudWatch, CloudFormation)

DevOps & CI/CD:

Docker, Jenkins, GitHub Actions, Git, Linux Administration, Automated Deployments

Software Engineering:

JavaScript, React, Node.js, Express.js, REST APIs, MERN Stack, HTML5, CSS3

Databases & Analytics:

MongoDB, Mongoose, PostHog (HogQL)

AI & Integrations:

OpenAI API, Gemini API, API Integration

Security:

OWASP Top 10, Web Application Security, Burp Suite, IAM Least Privilege

PROFESSIONAL EXPERIENCE

TOMASI | New York, USA | Remote

Lead Software Engineer & Cloud Architect | Full-Time | 2024 – Present

- **Owned the production technology stack end-to-end**, including application development, AWS infrastructure, deployments, monitoring and production operations.
- Designed and operated production AWS infrastructure for a live e-commerce platform using **ECS Fargate, ECR, ALB, CloudFront, Route 53 and CloudWatch**.
- Built automated **Docker-based CI/CD pipelines** using Jenkins and GitHub Actions to streamline repeatable production deployments.
- Developed and maintained the complete production e-commerce platform, including **authentication, payments, administration, inventory and analytics**.
- Engineered an event-driven **AWS Lambda + S3 serverless image optimisation pipeline** for automated image processing.
- Developed an analytics automation platform integrating **PostHog (HogQL), AWS and LLM APIs** to generate automated business insights.
- Applied **OWASP security practices, IAM least-privilege principles and performance optimisation** across the platform.

FEATURED PROJECTS**INTERACTIVE AWS SELF-HEALING INFRASTRUCTURE PLATFORM**

<https://arpanraj.space/>

- Built and deployed a **live AWS infrastructure demonstration** allowing users to intentionally terminate an ECS Fargate task and observe automatic service recovery.
- Designed an event-driven architecture using **API Gateway, Lambda, EventBridge, DynamoDB, ECS Fargate and CloudWatch**.
- Implemented reliability and cost controls using **ECS scheduling, Fargate Spot, scheduled scaling and AWS Budgets**.

- Protected infrastructure using **AWS WAF and IAM least-privilege access controls**.

TOMASI DESIGN — PRODUCTION E-COMMERCE PLATFORM

<https://tomasi.design/>

- Built and operated a production **MERN application** with authentication, inventory, administration, Stripe payments and responsive UI.
- Deployed application infrastructure using **Docker and AWS services**, integrating the application layer with production cloud infrastructure.
- **CI/CD** Single Pipeline using **Jenkins**

AI ANALYTICS & REPORTING AUTOMATION PLATFORM

- Built an automated analytics platform integrating **PostHog (HogQL), AWS and LLM APIs** to generate actionable business insights.
- Containerized and deployed services using **Docker, GitHub Actions and ECS Fargate**.

SERVERLESS IMAGE COMPRESSION PIPELINE

- Built an event-driven **AWS Lambda microservice triggered by S3 uploads** for automated image compression and processing.
- Designed the workflow around event-driven AWS services to minimize manual processing.

EDUCATION & CERTIFICATIONS

Bachelor of Computer Applications (BCA)

Navrachana University | **2026** | 7.87 CGPA

AWS Certified Cloud Practitioner

Amazon Web Services